

Chapter 32-STP States & Rapid Spanning Tree Protocol (RSTP)

1 STP Port Behavior Overview

Key Idea

✓ Root & Designated Ports → always **FORWARDING** (stable)

✗ Non-Designated Ports → always **BLOCKING** (stable)

Transitional States

Ports don't directly go from **blocking** → **forwarding**

👉 They pass through:

➡ Listening

➡ Learning

Total Delay in STP

Blocking → Listening (15 sec) → Learning (15 sec) → Forwarding

👉 Total ≈ **30 seconds delay** 🤖

Real Life Example

Think of **traffic signal** 🚦

- Red = Blocking
 - Yellow = Listening/Learning
 - Green = Forwarding
-

2 BLOCKING STATE (Stable)

What happens here?

-  No data traffic
-  No MAC learning
-  No forwarding

✓ Only prevents loops

Key Points

- Non-designated ports stay here
 - Interface is **disabled logically**
 - No normal communication
-

Important Concept

Initially:

-  All switch ports start in **blocking state**
 -  Then STP sends **BPDUs** to decide topology
-

3 LISTENING STATE (Transitional)

What happens here?

- ✓ Only BPDU send/receive
 -  No data traffic
 -  No MAC learning
-

Duration

 **15 seconds**

Key Points

- Only Root & Designated ports enter this
 - Non-designated ports remain blocking
-

Simple Understanding

Switch is:

 “Listening to network decisions”

LEARNING STATE (Transitional)

What happens here?

- ✓ Learns MAC addresses
 - ✓ Sends/receives BPDU
 - ✗ No data traffic
-

Duration

 **15 seconds**

Key Points

- Builds MAC table
 - Decides roles:
 - ✓ Root Bridge
 - ✓ Root Port
 - ✓ Designated Port
-

💡 Simple Understanding

Switch is:

👉 “Preparing to forward traffic”

🚀 5 FORWARDING STATE (Stable)

🟢 What happens here?

- ✓ Sends/receives data
 - ✓ Sends/receives BPDU
 - ✓ Learns MAC addresses
-

📌 Key Points

- Normal working state
 - Only Root & Designated ports reach here
-

🎯 Final Flow

Blocking → Listening → Learning → Forwarding

⚡ 6 STP TOOLKIT (Important Features)

🚀 PORTFAST

◆ What is PortFast?

👉 Makes port go directly to **FORWARDING**

▶ Skips:

- ✗ Listening
 - ✗ Learning
-

Where to Use?

✓ Only on **End Devices (PCs, Printers)**

Warning

✗ Never use on switch-to-switch link

👉 Can cause loops ⚡

Command

```
SW1(config)#interface g0/2  
SW1(config-if)#spanning-tree portfast
```

Real Life Example

Like:

👉 VIP lane at airport ✈️

No waiting, direct entry

BPDU GUARD

◆ What is BPDU Guard?

👉 Protects PortFast ports

If BPDU received:

✗ Port shuts down immediately

Command

```
SW1(config-if)#spanning-tree bpduguard enable
```

⚠️ Why Needed?

If someone connects a switch to PC port:

👉 Loop can occur

BPDU Guard:

👉 Stops this attack/security issue

🔥 Best Practice

Use together:

✓ PortFast + BPDU Guard

🌐 Global Commands

spanning-tree portfast default

spanning-tree portfast bpduguard default

🛡️ ROOT GUARD

◆ What is Root Guard?

👉 Prevents another switch from becoming root

🔧 Command

spanning-tree guard root

💡 Example

You don't want:

👉 Access switch becoming root

LOOP GUARD

◆ What is Loop Guard?

👉 Prevents loops when BPDU stops coming

Command

spanning-tree guard loop

💡 Use Case

If BPDU lost:

👉 Port should NOT become forwarding

⚡ 7 RAPID SPANNING TREE PROTOCOL (RSTP)

📌 Standard

👉 IEEE 802.1w

🚀 Why RSTP?

Classic STP:

🕒 ~30 sec delay

RSTP:

⚡ ~3 seconds

💡 Key Improvement

👉 No waiting for timers

👉 Uses handshake mechanism

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STP vs RSTP

Feature	STP (802.1D)	RSTP (802.1w)
Speed	Slow (~30 sec)	Fast (~3 sec)
BPDU	Only root sends	All switches send
Convergence	Timer-based	Event-based

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RSTP Port Costs

Speed	STP Cost	RSTP Cost
10 Mbps	100	2000000
100 Mbps	19	200000
1 Gbps	4	20000
10 Gbps	2	2000
100 Gbps	X	200

**10**

RSTP Port States

RSTP simplifies states:

State	Description
Discarding	Blocking + Disabled
Learning	Same
Forwarding	Same

Important

Discarding includes:

- ✓ Blocking
 - ✓ Shutdown
-



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RSTP Port Roles

Same as STP

- ✓ Root Port
 - ✓ Designated Port
-

New Roles



Alternate Port

- Backup to Root Port
 - If root fails → becomes forwarding
-



Backup Port

- Backup to Designated Port
 - Rare (used with HUB)
-



Built-in Features in RSTP

No need to configure:

- ✓ UplinkFast
 - ✓ BackboneFast
-

1 2 BPDU Behavior

STP

- Only Root sends BPDUs
 - Others forward
-

RSTP

✓ All switches send BPDUs

Failure Detection

Protocol Detection Time

STP 20 sec

RSTP 6 sec

1 3 RSTP Link Types

EDGE PORT

- ✓ Connected to PC
 - ✓ Works like PortFast
 - 👉 Direct forwarding
-

POINT-TO-POINT

- ✓ Switch to switch
 - ✓ Full duplex
-

SHARED

- ✓ Via HUB
- ✓ Half duplex

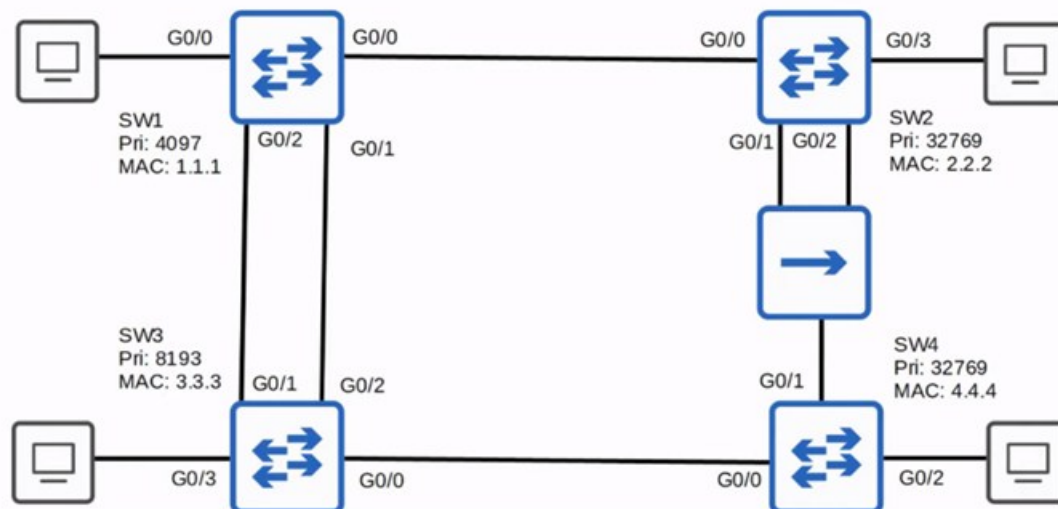
Commands

```
spanning-tree mode rapid-pvst  
show spanning-tree
```

Example Output Understanding

Gi0/0 → Designated → Forwarding
Gi0/1 → Backup → Blocking
Gi0/2 → Root → Forwarding

1 4 QUIZ – Final Topology Understanding



ROOT BRIDGE

- ✓ SW1 is Root Bridge

SW2

- G0/0 → Root Port
 - G0/1 → Designated
 - G0/2 → Backup
 - G0/3 → Designated
-

SW3

- G0/0 → Designated
 - G0/1 → Alternate
 - G0/2 → Root Port
 - G0/3 → Designated
-

SW4

- G0/1 → Alternate
 - G0/2 → Designated
-

Link Types

Connection Type

SW ↔ SW Point-to-Point

SW ↔ PC Edge

SW ↔ HUB Shared

Final Summary

STP

- ✓ Prevents loops
 - ✓ Uses blocking ports
 - ✓ Slow convergence
-

RSTP

- ✓ Faster (3 sec)
 - ✓ Better design
 - ✓ Uses alternate ports
-

Best Practice

- ✓ Use **Rapid PVST+** in real networks
-

Real Life Analogy

Think of:

 Office building elevators

- STP = normal elevator (slow wait)
 - RSTP = express elevator (fast direct)
-

Final Takeaway

- ✓ STP = Safety
 - ✓ RSTP = Speed + Safety
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1 Chapter Summary (Quick Revision)

STP Port Behavior

✓ Root Port & Designated Port → **Forwarding**

✗ Non-Designated Port → **Blocking**

STP State Flow

Blocking → Listening → Learning → Forwarding

 Total Delay ≈ **30 seconds** 

1. BLOCKING (Stable) ●

✗ No traffic

✗ No MAC learning

✓ Only BPDU

 Used to **prevent loops**

2. LISTENING (15 sec) ●

✓ BPDU exchange

✗ No traffic

✗ No MAC learning

 Switch is **deciding topology**

3. LEARNING (15 sec) ●

✓ MAC learning starts

✓ BPDU exchange

✗ No data forwarding

 Switch is **preparing**

4. FORWARDING (Stable) ●

- ✓ Full traffic
- ✓ MAC learning
- ✓ BPDU

 Normal operation

2 STP Toolkit (VERY IMPORTANT)

PortFast

 Directly goes to **Forwarding state**

- ✓ Skips delay
 - ✓ Used on end devices
 - ✗ Never use on switch links
-

BPDU Guard

 Protects PortFast ports

If BPDU received:

✗ Port shuts down

Root Guard

 Prevents other switches from becoming root

Loop Guard

 Prevents loops if BPDU stops

3 RSTP (Rapid STP)

 Standard: **IEEE 802.1w**

Why RSTP?

STP	RSTP
Slow (30 sec)	Fast (3 sec) ⚡
Timer-based	Event-based
Root sends BPDU	All switches send

RSTP States

State	Meaning
Discarding	Blocking + Disabled
Learning	Same
Forwarding	Same

RSTP Port Roles

Same as STP:

- ✓ Root Port
 - ✓ Designated Port
-

New Roles:

Alternate Port

- ✓ Backup to Root Port
 - ✓ Quickly becomes active
-

Backup Port

- ✓ Backup to Designated Port
 - ✓ Rarely used
-

RSTP Link Types

Type	Description
Edge	PC connection (like PortFast)
Point-to-Point	Switch-to-switch
Shared	Hub connection

RSTP Advantages

- ✓ Faster convergence (~3 sec)
 - ✓ No waiting timers
 - ✓ Better failure detection
 - ✓ Built-in optimizations
-

Chapter Conclusion

- 👉 STP ensures **network safety (loop prevention)**
- 👉 RSTP ensures **speed + fast recovery**

Without these:

- ✗ Network loops
- ✗ Broadcast storms
- ✗ Network downtime

With RSTP:

- ⚡ Fast failover
 - ⚡ Stable network
 - ⚡ Better performance
-

Q & A

1 What are STP states?

Answer

STP has 4 states:

- 1** Blocking
- 2** Listening
- 3** Learning
- 4** Forwarding

Each state ensures loop-free topology before forwarding traffic.

2 Why STP takes 30 seconds?

Answer

Because of:

-  Listening (15 sec)
-  Learning (15 sec)

Total = 30 sec delay

3 What happens in Blocking state?

Answer

- No traffic forwarding
- No MAC learning
- Only BPDU exchange

Used to prevent loops.

4 What is Learning state?

Answer

Switch starts learning MAC addresses but does not forward traffic.

5 What is PortFast?

Answer

PortFast allows a port to skip Listening and Learning states and go directly to Forwarding.

✓ Used for PCs

✗ Not used on switch links

6 What is BPDU Guard?

Answer

BPDU Guard shuts down a port if it receives BPDU.

Used to protect network from accidental loops.

7 What is Root Guard?

Answer

Prevents a switch from becoming root bridge.

Used to control network topology.

8 What is Loop Guard?

Answer

Prevents a port from becoming forwarding if BPDU stops.

Protects against unidirectional link failures.

9 What is RSTP?

Answer

Rapid Spanning Tree Protocol (RSTP) is an improved version of STP that provides faster convergence (~3 sec).

10 Difference between STP and RSTP?

Answer

Feature	STP	RSTP
Speed	Slow	Fast
BPDU	Root only	All switches
Convergence Timer		Event

1 1 What is Alternate Port?

Answer

Backup port to root port.

Immediately becomes forwarding if root port fails.

1 2 What is Edge Port?

Answer

Port connected to end device.

Works like PortFast.

1 3 What is Discarding state?

Answer

RSTP combines Blocking + Disabled into Discarding.

1 4 Why RSTP is faster?

Answer

Because:

- ✓ No timer wait
- ✓ Uses handshake mechanism
- ✓ Immediate convergence

1 5 Enable RSTP

spanning-tree mode rapid-pvst

1 6 Enable PortFast globally

spanning-tree portfast default

1 7 Enable BPDU Guard

spanning-tree portfast bpduguard default

1 8 Verify STP

show spanning-tree

SCENARIO QUESTIONS

1 9 Scenario

PC takes time to get network

Answer

Reason:

👉 STP delay (30 sec)

Solution:

✓ Enable PortFast

2 0 Scenario

Port suddenly shutdown

Answer

👉 BPDU Guard triggered

Reason:

Switch connected to PortFast port

2 1 Scenario

Network loop after adding switch

Answer

👉 PortFast misconfigured

Fix:

✓ Disable PortFast on trunk

2 2 Scenario

Link failure recovery slow

Answer

👉 Using STP

Solution:

✓ Use RSTP

2 3 Scenario

Unexpected root bridge change

Answer

👉 No Root Guard

Fix:

✓ Configure Root Guard

2 4 What is convergence?

Answer

Time taken for network to reach stable state after topology change.

2 5 STP vs RSTP failure detection?

Answer

Protocol Detection Time

STP 20 sec

RSTP 6 sec

2 6 What happens if BPDU is lost?

Answer

Without Loop Guard:

👉 Port may become forwarding → loop

With Loop Guard:

✓ Prevents loop

ONE-LINER

“STP prevents loops in a network, while RSTP improves convergence speed and ensures faster recovery from failures.”



FINAL REVISION POINTS

- ✓ STP has 4 states
 - ✓ Delay = 30 sec
 - ✓ PortFast removes delay
 - ✓ BPDU Guard protects network
 - ✓ RSTP = faster STP
 - ✓ RSTP uses alternate ports
 - ✓ Edge ports = fast forwarding
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